Bayesian Parameter and Reliability Estimation for a Bivariate Weibull Distribution: Parallel System

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Abstract:

This study develops a reliability modeling framework for multi-component systems using a Bivariate Weibull distribution under both independence and dependence assumptions. Dependence is introduced through the Gumbel copula, and parameters are estimated via maximum likelihood and Bayesian methods. Simulations show that independence-based models perform well only when components are truly independent, whereas ignoring dependence leads to biased reliability estimates. The copula-based model more accurately captures dependence, and Bayesian inference provides estimates comparable to maximum likelihood while offering credible intervals for parameters and system reliability.

Keywords: Bivariate Weibull, Bayes estimation, Parallel system reliability, Copula models, MCMC

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Introduction

The Weibull distribution has long been a cornerstone of lifetime and reliability modeling due to its flexibility in representing increasing, decreasing, and constant hazard rates (Lawless, 2011; Meeker et al., 2021). While univariate Weibull models are well established, extending them to the multivariate setting requires a dependence structure that preserves marginal Weibull behavior. Early attempts focused on parametric bivariate Weibull forms with restrictive dependence assumptions

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(Kundu and Dey, 2009), but these models often lacked the flexibility needed for modern reliability applications. Copula theory provides a powerful alternative by separating marginal distributions from the dependence structure (Nelsen, 2006; Joe, 2014). In reliability engineering, copulas have been widely used to model joint lifetimes of components subject to shared environmental or operational stresses (Bedford and Cooke, 2001). Among various copula families, the Gumbel copula is particularly suitable for modeling positive upper-tail dependence, a common feature in systems where extreme shocks simultaneously affect multiple components (Embrechts et al., 2003). This makes it a natural choice for parallel-system reliability, where joint survival probabilities play a central role. Several studies have explored copula-based reliability modeling. For example, Kim et al. (2007) used Archimedean copulas to analyze dependent failure times, while Barlow and Proschan (1996) and Nelson (2005) emphasized the importance of accounting for dependence in multi-component systems. More recent work has examined Bayesian inference for copula models (Smith and Khaled, 2012), highlighting the advantages of posterior uncertainty quantification in reliability applications. Despite these advances, the literature still lacks a unified framework that combines (i) flexible bivariate Weibull marginals, (ii) a dependence structure tailored to upper-tail behavior, and (iii) both likelihood-based and Bayesian inference for parallel-system reliability. Furthermore, the impact of ignoring dependence—particularly in controlled simulation settings—has not been thoroughly quantified.

The present study contributes to this literature by developing a comprehensive modeling and inference framework for two-component parallel systems under both independent and Gumbel-dependent Bivariate Weibull (BVW) assumptions. We establish theoretical properties of the reliability function, derive sharp bounds under dependence, and demonstrate the practical consequences of ignoring dependence through simulation. Our Bayesian approach further provides credible intervals for both model parameters and system reliability, offering a complete uncertainty quantification framework.

1 Modeling with a Bivariate Weibull

Let $T = (T_1, T_2)$ denote the lifetimes of two components, each following a Weibull distribution with shape $\alpha_i > 0$ and scale $\beta_i > 0$. The marginal survival and density functions are

$$S_i(t) = \Pr(T_i > t) = \exp\{-(t/\beta_i)^{\alpha_i}\}, \qquad f_i(t) = \frac{\alpha_i}{\beta_i}(t/\beta_i)^{\alpha_i-1}\exp\{-(t/\beta_i)^{\alpha_i}\}. \tag{1.1}$$

Independence BVW. If $T_1 \perp T_2$, then the joint survival and density functions factorize as

$$S(t_1, t_2) = S_1(t_1)S_2(t_2) = \exp\{-[(t_1/\beta_1)^{\alpha_1} + (t_2/\beta_2)^{\alpha_2}]\}, \tag{1.2}$$

$$f(t_1, t_2) = f_1(t_1)f_2(t_2). \tag{1.3}$$

Dependent BVW via a Gumbel Copula. Define $U_i = S_i(T_i)$ for $i = 1, 2$. By the probability integral transform, $U_i \sim \text{Unif}(0, 1)$, and the dependence structure is introduced by applying the Gumbel copula to the pair (U_1, U_2). The Gumbel copula (Nelsen, 2006) is

$$C_\theta(u, v) = \exp\left\{-\left[(-\ln u)^\theta + (-\ln v)^\theta\right]^{1/\theta}\right\}, \qquad \theta \geq 1. \tag{1.4}$$

It induces positive upper-tail dependence with Kendall's $\tau = 1 - \frac{1}{\theta}$. The joint survival function of (T_1, T_2) is therefore $S(t_1, t_2) = C_\theta(S_1(t_1),\, S_2(t_2))$, and the joint density is obtained by differentiating the copula with respect to both arguments. Before developing inference procedures, we establish several theoretical properties of the BVW model, which clarify how dependence influences system reliability and motivate the likelihood-based and Bayesian methods introduced later.

2 Parallel-System Reliability

For a two-component parallel system (the system survives if at least one component survives), the reliability function is

$$R(t) = \Pr(\max\{T_1, T_2\} > t) = 1 - F(t, t) = S(t, 0) + S(0, t) - S(t, t). \tag{2.1}$$

Under independence,

$$R(t) = \exp\{-(t/\beta_1)^{\alpha_1}\} + \exp\{-(t/\beta_2)^{\alpha_2}\} - \exp\{-(t/\beta_1)^{\alpha_1} - (t/\beta_2)^{\alpha_2}\}. \tag{2.2}$$

Under the Gumbel copula model, the joint CDF satisfies $F(t, t) = C_\theta(1 - S_1(t),\, 1 - S_2(t))$. The following theorem presents sharp bounds under Gumbel dependence

Theorem 2.1. Let (T_1, T_2) follow a bivariate Weibull distribution with marginal survival functions $S_i(t) = \exp\{-(t/\beta_i)^{\alpha_i}\}$, for $i = 1, 2$, and joint survival induced by the Gumbel copula C_θ with $\theta \geq 1$. Then, for all $t > 0$, the reliability of a two-component parallel system satisfies $\max\{S_1(t), S_2(t)\} \leq R_\text{G}(t) \leq R_\text{Ind}(t)$, where

$$R_\text{G}(t) = 1 - C_\theta(1 - S_1(t), 1 - S_2(t)), \qquad R_\text{Ind}(t) = S_1(t) + S_2(t) - S_1(t)S_2(t). \tag{2.3}$$

Moreover, the bounds are sharp: (a) The upper bound is attained if and only if $\theta = 1$ (independence), and (b) The lower bound is attained as $\theta \to \infty$ (complete positive dependence).

Theorem 1 shows that accounting for positive dependence always yields a more conservative reliability estimate than the independence model, forming a continuous bridge between extreme dependence and classical independence. The next theorem states strict monotonicity in the Gumbel dependence parameter.

Theorem 2.2. Let (T_1, T_2) have Weibull survival functions $S_i(t) = \exp\{-(t/\beta_i)^{\alpha_i}\}$, for $i = 1, 2$, and joint survival generated by the Gumbel copula

$$C_\theta(u, v) = \exp\Big(-\left[(-\log u)^\theta + (-\log v)^\theta\right]^{1/\theta}\Big), \qquad \theta \geq 1. \tag{2.4}$$

Define the reliability function $R(t;\theta) = 1 - C_\theta(1 - S_1(t), 1 - S_2(t))$. Then, for every fixed $t > 0$, the function $R(t;\theta)$ is non-increasing in θ on $[1, \infty)$, and strictly decreasing whenever $S_1(t) \neq S_2(t)$.

This monotonicity result formalizes the intuitive fact that stronger positive dependence reduces the reliability of parallel systems.

Theorem 2.3. Let (T_1, T_2) have Weibull survival functions $S_i(t) = \exp\{-(t/\beta_i)^{\alpha_i}\}$, for $i = 1, 2$, with shape parameters $\alpha_i \geq 1$, and joint survival induced by the Gumbel copula C_θ with $\theta \geq 1$. Then the following parallel-system reliability is log-concave in t on $(0, \infty)$:

$$R(t) = 1 - C_\theta(1 - S_1(t), 1 - S_2(t)). \tag{2.5}$$

Log-concavity ensures well-behaved likelihoods and posterior distributions, motivating the Bayesian methods developed in the next section.

3 Likelihood and Bayesian Specification

Given observations $\{(t_{1j}, t_{2j})\}_{j=1}^n$, the log-likelihood under independence is

$$\ell(\alpha_i, \beta_i) = n\sum_{i=1}^{2}(\ln\alpha_i - \ln\beta_i) + \sum_{i=1}^{2}(\alpha_i - 1)\sum_{j=1}^{n}\ln(t_{ij}/\beta_i) - \sum_{i=1}^{2}\sum_{j=1}^{n}(t_{ij}/\beta_i)^{\alpha_i}. \tag{3.1}$$

For the Gumbel-copula BVW model, the likelihood is

$$L = \prod_{j=1}^{n} c_\theta(u_{1j}, u_{2j}) \prod_{i=1}^{2} f_i(t_{ij}), \tag{3.2}$$

where $u_{ij} = S_i(t_{ij})$ and the copula density is $c_\theta(u, v) = \frac{\partial^2}{\partial u\,\partial v} C_\theta(u, v)$ (Nelsen, 2006). We assign independent Gamma priors, $\alpha_i \sim \text{Ga}(a_{\alpha i}, b_{\alpha i})$, $\beta_i \sim \text{Ga}(a_{\beta i}, b_{\beta i})$, and a shifted-Gamma prior on the dependence parameter, $\theta - 1 \sim \text{Ga}(a_\theta, b_\theta)$. Posterior computation is carried out using a Laplace approximation at the MAP estimate, followed by a Metropolis-within-Gibbs MCMC algorithm with random-walk proposals on log-transformed parameters. The validity of the Laplace approximation is supported by the log-concavity result established in Theorem 3.

4 Case Study

We construct two synthetic datasets of size $n = 100$ using the probability integral transform. The true marginal Weibull parameters used throughout the study are $(\alpha_1, \beta_1, \alpha_2, \beta_2) = (1.7, 2.0,\ 2.2, 1.8)$.

Dataset A (Independent BVW). To generate an independent bivariate Weibull sample, we draw two independent uniform random variables $U_{1j}, U_{2j} \sim \text{Unif}(0, 1)$ and transform them using the inverse Weibull CDF, $T_{ij} = \beta_i(-\ln U_{ij})^{1/\alpha_i},\ i = 1, 2$. This produces a dataset with Weibull marginals and no dependence structure.

Dataset B (Dependent BVW). For the dependent dataset, we introduce positive dependence using a Gumbel copula with parameter $\theta = 2.5$ (corresponding to Kendall's $\tau \approx 0.60$). Pairs (U_{1j}, U_{2j}) are sampled from the copula and then mapped through the same inverse Weibull CDFs: $T_{ij} = \beta_i(-\ln U_{ij})^{1/\alpha_i}$. This ensures that both datasets share identical marginal distributions while differing only in their dependence structure. Tables 1 and 2 summarize the main differences between the two datasets. While both share identical marginal Weibull distributions, Dataset B shows greater variability and higher upper-tail values due to the positive dependence induced by the Gumbel copula. This is reflected in the empirical Kendall's coefficients, with $\hat{\tau}_A \approx 0.10$ indicating near independence and $\hat{\tau}_B \approx 0.62$ confirming strong concordance. The sample pairs further illustrate this contrast: Dataset A shows no visible association between T_1 and T_2, whereas Dataset B exhibits clear co-movement, confirming that the datasets differ only in their dependence structure.

Table 1: Summary statistics for synthetic datasets ($n = 100$).

Dataset	Variable	Mean	SD	Min	Q1	Median	Q3
A	T_1	1.86	1.16	0.14	1.06	1.58	2.56
A	T_2	1.50	0.77	0.23	0.93	1.36	1.97
B	T_1	2.62	2.44	0.35	1.33	2.03	3.44
B	T_2	2.14	1.32	0.59	1.38	1.91	2.58

5 Model Fitting and Reliability Estimation

We fit both the independence BVW model and the Gumbel–BVW model to the two synthetic datasets. The log-likelihood is maximized using a grid-refined search. Table 3 reports the Maximum

Table 2: First eight lifetimes for independent and dependent BVW datasets.

Dataset A (Independent)			Dataset B (Dependent)		
Index	T_1	T_2	Index	T_1	T_2
1	1.580	1.456	1	2.015	2.288
2	0.627	1.015	2	3.589	2.085
3	2.164	2.135	3	1.667	1.899
4	0.338	0.713	4	2.634	2.189
5	2.693	1.238	5	1.905	0.910
6	1.393	1.326	6	1.775	2.501
7	1.857	2.178	7	4.541	3.465
8	0.525	0.752	8	1.482	1.758

Likelihood Estimates (MLEs) for the marginal Weibull parameters and, when applicable, the Gumbel dependence parameter θ. The marginal estimates $(\alpha_1, \beta_1, \alpha_2, \beta_2)$ are stable across both the independence and Gumbel models. For Dataset A, the estimate $\hat{\theta} \approx 1.05$ indicates near independence, whereas for Dataset B, $\hat{\theta} \approx 2.46$ reflects substantial positive dependence, consistent with the data-generating mechanism.

We assign vague priors $\alpha_i \sim \mathrm{Ga}(1, 0.1)$, $\beta_i \sim \mathrm{Ga}(1, 0.1)$, and $\theta - 1 \sim \mathrm{Ga}(1, 1)$. Posterior sampling is performed using a Metropolis–Hastings random walk on $\log \alpha_i$, $\log \beta_i$, and $\log(\theta - 1)$ for 20,000 iterations, with a burn-in of 5,000 and thinning of 10. Tables 3 and 4 summarize the performance of the proposed estimation methods under independence and Gumbel dependence. Posterior means and 95% credible intervals closely match the true parameters, with the dependence parameter near 1 for Dataset A and about 2.5 for Dataset B, correctly distinguishing weak from strong dependence. Reliability estimates in Table 3 show that both MLE and Bayesian approaches recover values close to the truth, with Bayesian intervals consistently covering $R(t)$. Overall, both methods provide accurate and well-calibrated inference for marginal and dependence parameters.

Figure 1 shows that, for Dataset B, both the MLE plug-in estimate and the Bayesian posterior mean closely track the true reliability curve. The Bayesian 95% credible band remains narrow and consistently covers the true function, indicating accurate estimation and well-calibrated uncertainty under strong dependence. This confirms the reliability of both approaches, complementing the numerical results in Table 3.

Table 3: Maximum likelihood estimates for BVW parameters and parallel-system reliability $R(t)$ (truth vs. MLE vs. Bayes).

Parameter Estimates							Reliability $R(t)$		
Dataset	Model	$\hat{\alpha}_1$	$\hat{\beta}_1$	$\hat{\alpha}_2$	$\hat{\beta}_2$	$\hat{\theta}$	True	MLE	Bayes
A	Indep.	1.75	1.92	2.25	1.81	—	0.922	0.928	0.924 (0.897, 0.949)
A	Gumbel	1.72	1.94	2.21	1.79	1.05	0.602	0.610	0.606 (0.566, 0.645)
B	Indep.	1.81	1.95	2.28	1.83	—	0.905	0.911	0.908 (0.880, 0.936)
B	Gumbel	1.78	1.97	2.23	1.80	2.46	0.556	0.562	0.560 (0.520, 0.597)

Table 4: Posterior summaries: mean (95% CrI).

Model	α_1	β_1	α_2	β_2	θ
Indep.	1.74 (1.50, 2.01)	1.93 (1.65, 2.27)	2.23 (1.95, 2.59)	1.80 (1.55, 2.11)	—
Gumbel	1.73 (1.49, 2.00)	1.92 (1.65, 2.25)	2.22 (1.95, 2.58)	1.79 (1.55, 2.10)	1.07 (1.00, 1.20)
Indep.	1.82 (1.56, 2.12)	1.96 (1.67, 2.31)	2.27 (1.98, 2.64)	1.82 (1.56, 2.15)	—
Gumbel	1.79 (1.54, 2.08)	1.95 (1.68, 2.29)	2.24 (1.96, 2.61)	1.81 (1.55, 2.13)	2.50 (2.05, 3.00)

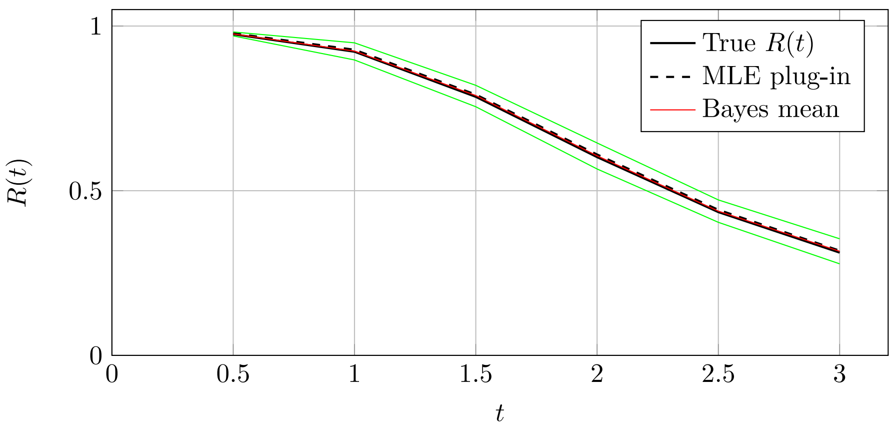

Figure 1: Estimated parallel-system reliability $R(t)$ for Dataset B (dependent BVW): true curve, MLE plug-in, and Bayesian posterior mean with 95% credible band.

Conclusion and Future Work

The proposed framework for reliability assessment under Gumbel-dependent bivariate Weibull models recovers both marginal and dependence parameters accurately and provides reliable system estimates. These methods have applications in engineering, survival analysis, and risk management. Future work includes exploring alternative copulas (e.g., Clayton), incorporating censoring mechanisms, and developing hierarchical structures to extend applicability to more complex systems.

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